

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO1	Assessment:	Assessment Tool 3 – Experiments
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.		
Student Name:	CH. Durga Prasad	Reg. No.:	192425305
Section / Batch:	AIML	Date:	

Code of Conduct

I CH. Durga Prasad certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules

Name: - CH. Durga Prasad

Course code: - MLA0407

Reg no: - 192425305

Course Name: - Deep Learning

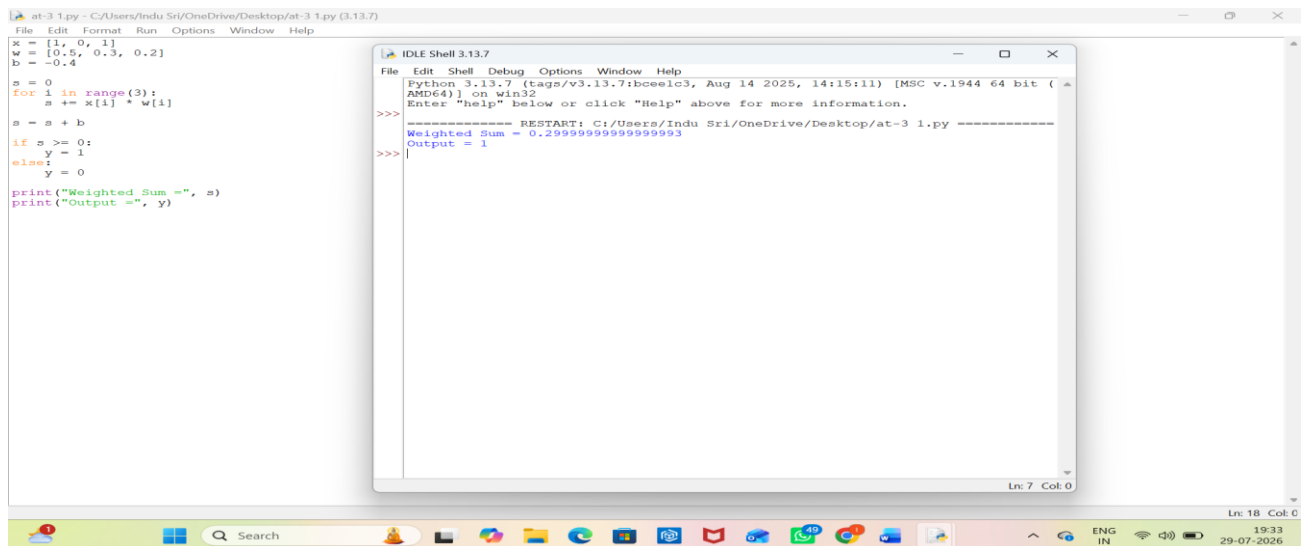
Experiment 1: Implementation of a Single Artificial Neuron Objective: Implement a single artificial neuron using Python. Task: Total Marks: 50 Annexure A Experiments

Aim

To implement a single artificial neuron using Python that accepts multiple inputs, computes the weighted sum, applies a step activation function, and generates the output.

Algorithm

1. Start.
2. Define input values.
3. Define weights and bias.
4. Calculate weighted sum.
5. Apply step activation function.
6. Display output.
7. Stop.



```
at-3 1.py - C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py (3.13.7)
File Edit Format Run Options Window Help
x = [1, 0, 1]
w = [0.5, 0.3, 0.2]
b = -0.4

s = 0
for i in range(3):
    s += x[i] * w[i]

s = s + b
if s >= 0:
    y = 1
else:
    y = 0

print("Weighted Sum =", s)
print("Output =", y)
```

```
IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ----- RESTART: C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py -----
>>> Weighted Sum = 0.29999999999999993
>>> Output = 1
>>> |
```

Result: -

The artificial neuron successfully computed the weighted sum and produced the correct binary output using the step activation function.

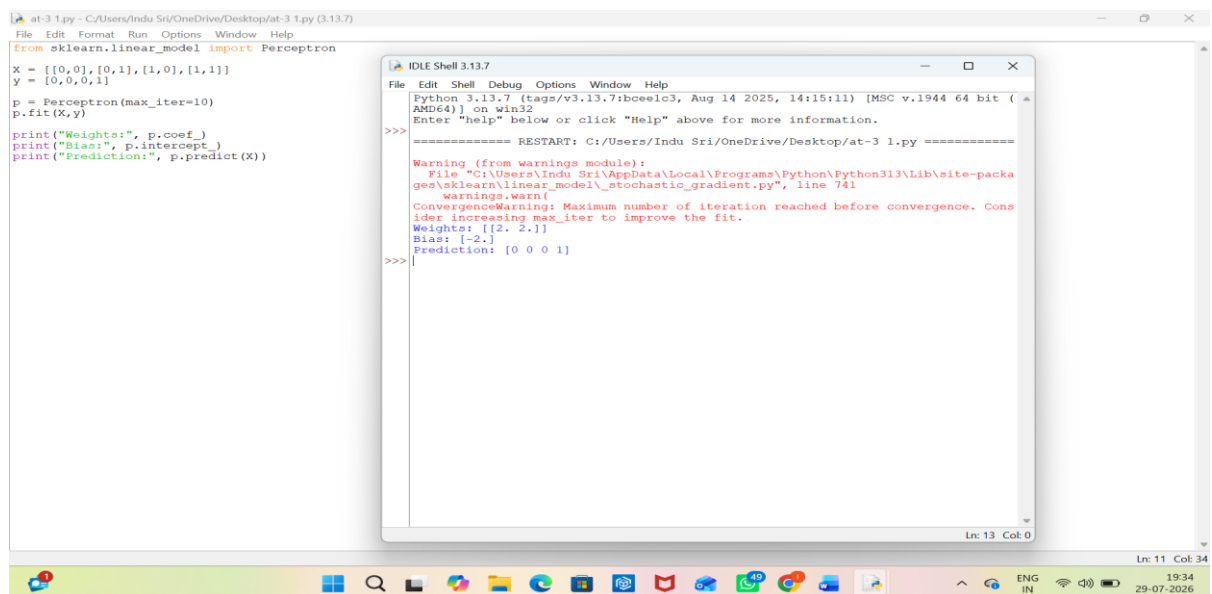
Experiment 2: Perceptron for AND Gate Objective: Implement the Perceptron Learning Algorithm. Task:

Aim

To implement the Perceptron Learning Algorithm for the AND logic gate.

Algorithm

1. Start.
2. Define AND gate inputs and outputs.
3. Initialize weights and bias.
4. Train the perceptron.
5. Update weights using the perceptron rule.
6. Display final weights and predictions.
7. Stop.



```
at-3 1.py - C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py (3.13.7)
File Edit Format Run Options Window Help
from sklearn.linear_model import Perceptron

x = [[0,0],[0,1],[1,0],[1,1]]
y = [0,0,0,1]

p = Perceptron(max_iter=10)
p.fit(x,y)

print("Weights:", p.coef_)
print("Bias:", p.intercept_)
print("Prediction:", p.predict(x))

IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceec13, Aug 14 2025, 14:15:11) [MSC v.1914 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py =====
Warning (from warnings module):
  File "C:/Users/Indu Sri/AppData/Local/Programs/Python/Python313/Lib/site-packages/sklearn/linear_model/_stochastic_gradient.py", line 741:
    warnings.warn(
ConvergenceWarning: Maximum number of iteration reached before convergence. Consider increasing max_iter to improve the fit.
Weights: [[2. 2.]]
Bias: [-2.]
Prediction: [0 0 0 1]
>>>
```

Result: -

The perceptron correctly classified the AND gate with 100% accuracy.

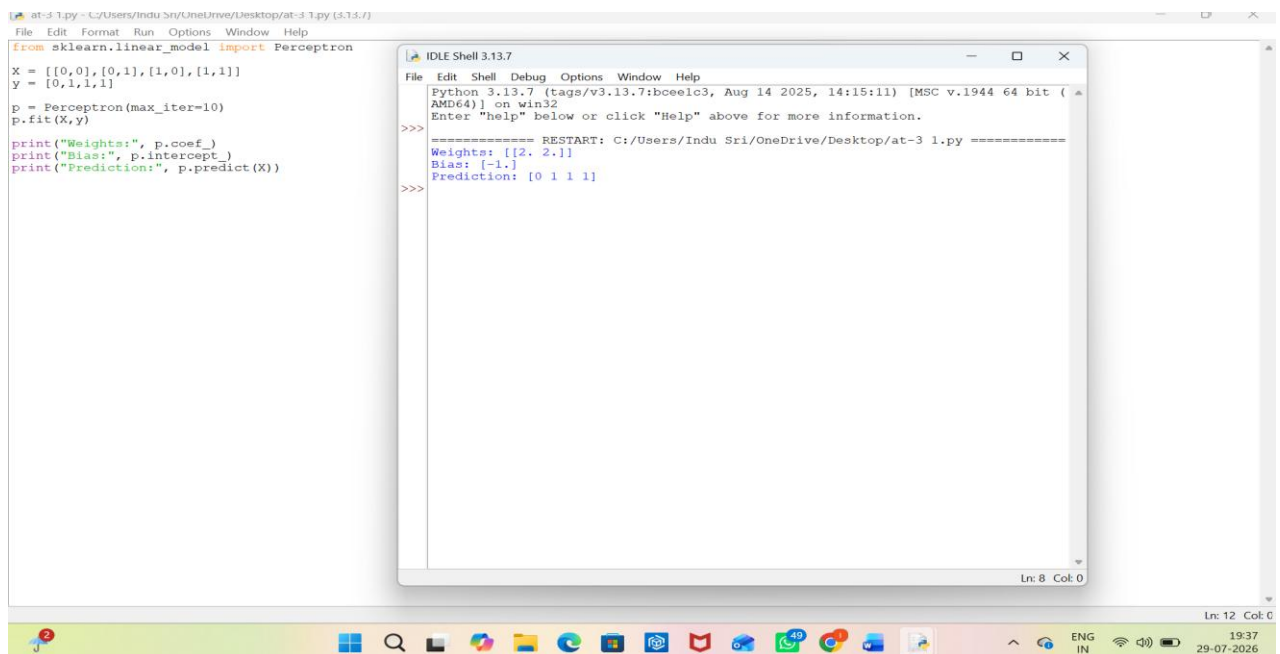
Experiment 3: Perceptron for OR Gate Objective: Design a perceptron model for binary classification. Task: Implement a perceptron to classify the OR logic gate.

Aim

To implement a perceptron model for the OR logic gate.

Algorithm

1. Start.
2. Define OR gate inputs and outputs.
3. Initialize weights and bias.
4. Train the perceptron.
5. Predict outputs.
6. Display weights and predictions.
7. Stop.



```
at-3 1.py - C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py (3.13.7)
File Edit Format Run Options Window Help
from sklearn.linear_model import Perceptron
X = [[0,0],[0,1],[1,0],[1,1]]
y = [0,1,1,1]
p = Perceptron(max_iter=10)
p.fit(X,y)
print("Weights:", p.coef_)
print("Bias:", p.intercept_)
print("Prediction:", p.predict(X))

IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py =====
Weights: [[2. 2.]]
Bias: [-1.]
Prediction: [0 1 1 1]
>>>
```

Result: -

The perceptron correctly classified the OR gate with **100% accuracy**.

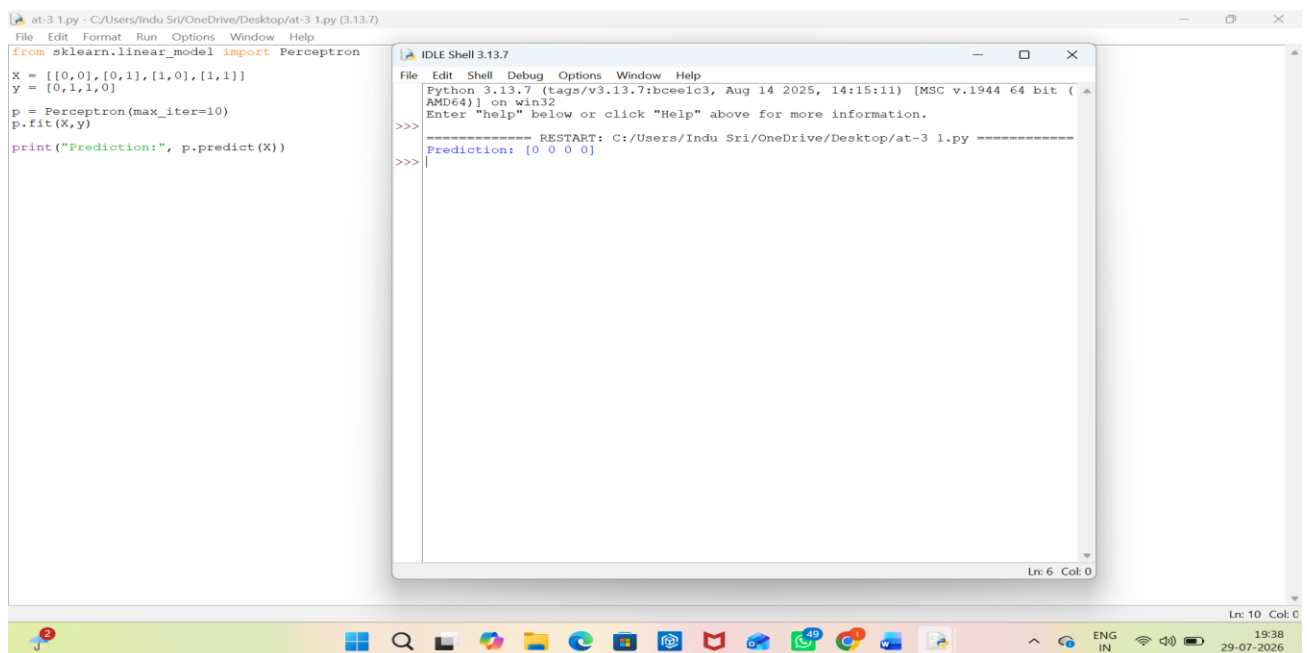
Experiment 4: XOR Classification Analysis Objective: Study the limitations of a single-layer perceptron. Task: Attempt to train a perceptron for the XOR logic gate.

Aim

To study the limitations of a single-layer perceptron using the XOR logic gate.

Algorithm

1. Start.
2. Define XOR inputs and outputs.
3. Train a perceptron.
4. Predict outputs.
5. Observe incorrect classification.
6. Conclude that XOR is not linearly separable.
7. Stop.



```
at-3 1.py - C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py (3.13.7)
File Edit Format Run Options Window Help
from sklearn.linear_model import Perceptron
X = [[0,0],[0,1],[1,0],[1,1]]
y = [0,1,1,0]
p = Perceptron(max_iter=10)
p.fit(X,y)
print("Prediction:", p.predict(X))

IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceec1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py =====
Prediction: [0 0 0 0]
>>>
```

Result: -

The single-layer perceptron failed to classify the XOR gate because it is not linearly separable.

Experiment 5: Gradient Descent Optimization Objective: Understand Gradient Descent.

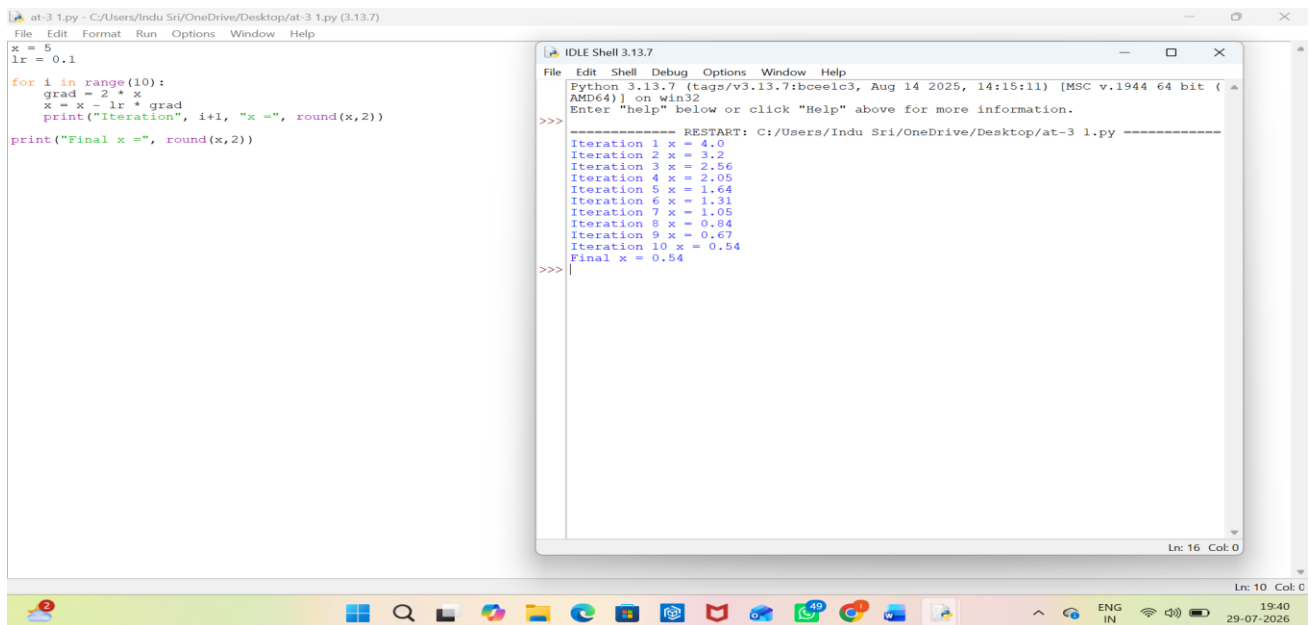
Task: Implement the Gradient Descent algorithm to minimize a simple quadratic cost function.

Aim

To implement the Gradient Descent algorithm for minimizing a quadratic cost function.

Algorithm

1. Start.
2. Initialize learning rate and parameter.
3. Compute gradient.
4. Update parameter using Gradient Descent.
5. Repeat until convergence.
6. Plot cost versus iterations.
7. Stop.



```
at-3 1.py - C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py (3.13.7)
File Edit Format Run Options Window Help
x = 5
lr = 0.1
for i in range(10):
    grad = 2 * x
    x = x - lr * grad
    print("iteration", i+1, "x =", round(x,2))
print("Final x =", round(x,2))

IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
----- RESTART: C:/Users/Indu Sri/OneDrive/Desktop/at-3 1.py -----
Iteration 1 x = 4.0
Iteration 2 x = 3.2
Iteration 3 x = 2.56
Iteration 4 x = 2.05
Iteration 5 x = 1.64
Iteration 6 x = 1.31
Iteration 7 x = 1.05
Iteration 8 x = 0.84
Iteration 9 x = 0.67
Iteration 10 x = 0.54
Final x = 0.54
>>>
```

Result: -

Gradient Descent successfully minimized the cost function, showing convergence over the iterations.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____